
COURSE TOPICS

1. **Introduction:** The pre-standard PLC landscape, what the five-language specification unified, and what it deliberately left vendor-specific.
2. **Data Types:** BOOL, INT, DINT, REAL, TIME, STRING, and user-defined structures; type mismatch consequences in live systems.
3. **Structured Text:** Pascal-style syntax — IF/THEN, FOR, WHILE, CASE — and appropriate use cases among the five languages.
4. **Ladder Diagram:** Contact/coil notation, rung structure, and reading existing automation code in the industry's most common language.
5. **Function Block Diagram:** Data-flow graphical programming, block connectivity, and use cases in PID and signal conditioning.
6. **Sequential Function Chart:** Steps, transitions, and action qualifiers for modeling processes with discrete ordered states.
7. **Program Organization Units:** Functions, Function Blocks, Programs, and Configurations; reusable library design across vendors.
8. **RTAC Integration:** IEC 61131-3 in ACSELERATOR Architect and translation from PLC programming to substation automation logic.
9. **Troubleshooting:** Implicit type conversion, scan cycle overruns, latching coil failures, and silent runtime faults.
10. **Practice Lab:** PID controller in Structured Text, motor starter in Ladder, batch sequence in SFC, analog scaling Function Block.

LEARNING OUTCOMES

- Write and debug programs in Structured Text, Ladder Diagram, and Sequential Function Chart.
- Design reusable Function Blocks with typed interfaces for use across multiple vendor platforms.
- Interpret and modify existing Ladder Diagram code in a production automation system.
- Configure a PID controller using the standard IEC 61131-3 function block interface.
- Diagnose scan cycle overruns, type conversion errors, and silent runtime faults in live PLC programs.

Certification Relevance

- **PLCopen Certified** — Certification exams test IEC 61131-3 language knowledge, POU's, and programming best practices directly.
- **ISA CCST** — PLC programming, ladder logic, and control system configuration are core tested topics at all levels.
- **Vendor Credentials** — Siemens TIA Portal, Rockwell Studio 5000, and Beckhoff TwinCAT all use IEC 61131-3 as their programming foundation.

Next Steps

1. Take IEC 61131-3 before the RTAC guide. The logic concepts transfer directly to ACSELERATOR Architect.
2. Download CODESYS (free, IEC 61131-3 compliant IDE) and run the lab exercises in a soft PLC.
3. Review the PLCopen Function Block libraries for motion, communications, and safety.
4. PLCopen certification or a vendor credential (TIA Portal, Studio 5000) is the right first PLC-specific credential.